Cart lab: change mass

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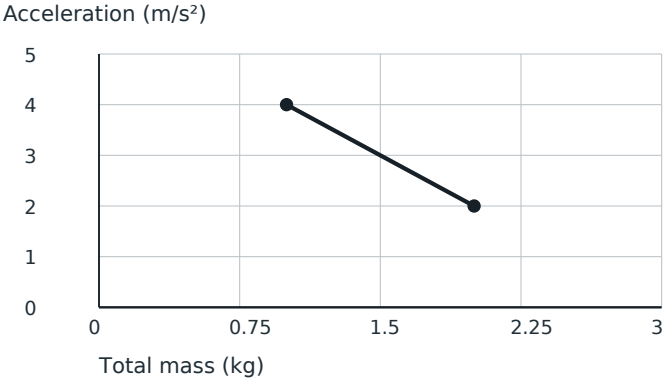
Learning target: Use a controlled comparison to describe the effect of system mass on acceleration.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 8.*
a=3 m/s² east; average velocity 6 m/s east; displacement 24 m east.

02. Worked example: explain the model
Points (1, 4),(2, 2); constant net force 4 N. Two points alone do not establish every detail of a curve.

Model / diagram check



03. Read the sample pattern
Mass doubles from 1 to 2 kg, while acceleration halves from 4 to 2 m/s². Both trials give ma = 4 N.

04. Plan a fair test
Change total mass by adding to cart; measure acceleration; keep hanger mass/track/release same. More hanger changes driving force too.

05. Record evidence honestly
Teacher checks measured values and repeatability if collected. Supplied comparison 1 kg→2 kg: 4→2 m/s². Never claim supplied numbers are measured.

06. Review: connect an earlier idea *Review: Unit 1, Day 10.*
Earth pulls the sign down with 40 N; the cord pulls it up with 40 N.

07. Exit check
3 m/s²; same net force, comparable resistance. In real data deviations require checking controls.

08. STRETCH: test the model
1 m/s²; keep driving/friction conditions controlled, measure actual acceleration and compare including uncertainty.

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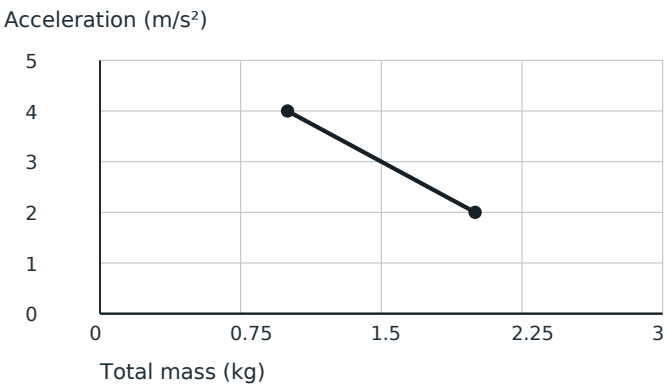
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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 8.*
Stops after 6 s at distance 72 m. Extending to 8 s predicts -8 m/s, which assumes continued acceleration after stopping and reversal.

02. Worked example: explain the model
Points (1, 4),(2, 2); constant net force 4 N. Two points alone do not establish every detail of a curve.

Model / diagram check



03. Test an inverse relationship
1/M=2, 1.333, 1, 0.667 kg^{-1}; straight through origin with slope≈0.49 N, the net force.

04. System boundary
Hanger accelerates with cart; tension forces between components cancel in combined system. Idealizations massless string/pulley, negligible friction, inextensible string.

05. Measured slope and repetitions
a=0.30/0.60=0.50 m/s^2. Table includes M, 3 trials, mean a, 1/M with units. No invented measurement accepted.

06. Review: connect an earlier idea *Review: Unit 1, Day 10.*
Weight 29.4 N down from Earth and tension 29.4 N up from string. Object-on-string acts on the string, not the object.

07. Exit check
F=0.48 N; a=0.40 m/s^2. Greater resistance lowers net force and observed acceleration.

08. STRETCH: test the model
Accelerations 6, 3, 1.5 m/s^2. Plot a versus 1/m; slope 3 N. Net force must be controlled, not assumed from a drive setting alone.